

Model Justification and Literature Integration

This document explains why we made each modelling decision and which academic sources support those decisions.

Introduction

The Vehicle Routing Problem (VRP) is one of the most extensively studied optimisation problems in Operations Research. Since its introduction by Dantzig and Ramser (1959), numerous variants have been developed to represent increasingly realistic transportation systems. Among these, the Capacitated Vehicle Routing Problem (CVRP) extends the classical VRP by introducing vehicle capacity limitations, while later extensions incorporate heterogeneous fleets, customer time windows and time-dependent travel times.

The objective of this document is to justify the modelling decisions adopted in the Dr. Oetker–Lidl distribution project. Rather than selecting modelling assumptions arbitrarily, every component of the optimisation model is supported by established literature in the Vehicle Routing Problem domain.

Consequently, this document explains how the classical CVRP was extended into a Time-Dependent Heterogeneous Vehicle Routing Problem with Time Windows (TDHVRPTW) in order to accurately represent the operational characteristics of the case study.

1. Why is this problem a TDHVRPTW?

Our problem involves delivering pallets from one depot to 25 stores using a mixed fleet of trucks, where each store must be reached within a specific time slot and travel times change depending on traffic during the day.

This matches exactly the definition of the Time-Dependent Heterogeneous Vehicle Routing Problem with Time Windows (TDHVRPTW) as described in the literature.

The basic VRP was introduced by Dantzig and Ramser (1959) as the problem of routing vehicles from a central depot to a set of customers at minimum cost. Our problem extends this in three directions:

First, our fleet is heterogeneous. Toth and Vigo (2014) define the Heterogeneous Fleet VRP as the variant in which vehicles differ in capacity and, in some cases, operating cost. Such fleet structures are common in industrial logistics because they provide greater flexibility than a homogeneous fleet. Since the Dr. Oetker distribution network operates heavy trucks with 33 pallets and medium trucks with 12 pallets, where a medium truck cannot take over a heavy truck's route without violating its capacity, the heterogeneous fleet assumption was adopted in the present study.

Second, our stores have time windows. All 25 Lidl stores must receive their delivery between 08:00 and 12:00. This is the hard time window constraint described by Solomon (1987), who showed that adding time windows fundamentally changes the structure of the routing problem and makes it significantly harder to solve.

Third, travel times in our network are not constant. The OWL road network experiences morning rush hour congestion between 07:00 and 09:00 that increases travel times by approximately 30%. Treating travel times as static in a congested network produces routes that look optimal on paper but arrive late in practice, which is why a simple CVRP cannot represent our problem.

Combining all three extensions gives us the TDHVRPTW, which is the minimum model that accurately represents the Dr. Oetker distribution problem.

2. Why did we choose these constraints?

The mathematical formulation incorporates several operational constraints that ensure the generated routes remain both physically feasible and operationally realistic. These constraints are derived from the characteristics of the Dr. Oetker distribution system and are consistent with established Vehicle Routing Problem literature.

Capacity constraint (C4)

Each vehicle route must not carry more pallets than the truck capacity. This is the fundamental constraint of the CVRP introduced by Clarke and Wright (1964). Without this constraint the model would assign unlimited pallets to a single truck which is physically impossible.

We use two capacity levels: $Q_{\text{heavy}} = 33$ pallets and $Q_{\text{medium}} = 12$ pallets. These values come from standard European truck configurations for regional food distribution.

Time window constraint (C6)

All stores must be served between 08:00 and 12:00. This reflects the real operational policy of Lidl stores in the OWL region where receiving staff and loading docks are only available during this window. Solomon (1987) shows that hard time windows are appropriate when the customer cannot accept early or late delivery under any circumstances.

Maximum route duration (C7)

No route may exceed 405 minutes. This comes from German labour law which requires a 45 minute break after 6 hours of driving. A standard morning shift of 7 hours minus the break and a 30 minute traffic buffer gives 405 minutes of effective route time.

Time-dependent travel times (C5)

We apply a congestion multiplier $\mu(t)$ to all travel times based on time of day. The values we use are:

- 07:00 to 09:00: multiply by 1.3
- 09:00 to 11:30: multiply by 1.0
- 11:30 to 12:00: multiply by 1.1

A step function is a practical and commonly used way to approximate congestion in urban and regional networks when continuous, sensor-level traffic data is not available, and this kind of time-of-day approximation is a standard simplification in time-dependent routing models.

Service time formula

We use $s_i = 10 + 2 \times q_i$ minutes at each store. The 10 minute fixed component covers parking, driver check-in, and paperwork. The 2 minutes per pallet covers physical unloading. This follows the common modelling practice in the VRP literature of using a linear, demand-dependent service time in heterogeneous fleet models.

3. Why did we use Google Maps for the distance and time matrices?

The TDHVRPTW requires real road distances and real travel times. Using Euclidean straight-line distances would underestimate actual driving distances because trucks must follow roads, respect one-way streets, and use motorways or regional roads depending on the destination.

For this problem a 5 minute error in travel time per arc multiplied across 25 stores means routes that look feasible in the model may arrive late in reality. Since our time windows are hard constraints with only a 4 hour delivery window, accuracy matters.

Google Maps Distance Matrix API provides actual driving distances and free-flow travel times between any two locations using the real road network. We used it to build a complete 26 by 26 matrix covering the depot and all 25 stores. This approach follows the general practice in applied VRP studies of using real road network data rather than synthetic instances.

The use of actual road-network distances improves the external validity of the case study because optimisation results better represent practical transportation conditions than synthetic benchmark instances.

4. Why did we choose Clarke-Wright and OR-Tools?

Clarke-Wright Savings Algorithm

Clarke and Wright (1964) introduced the savings algorithm as an efficient constructive heuristic for the VRP. It works by computing a saving value for every pair of customers that measures how much distance is saved by combining them on one route instead of serving each with a separate vehicle.

We chose Clarke-Wright because it is the most cited and most studied constructive heuristic for the VRP family. It provides a transparent and interpretable solution method that allows us to explain step by step how routes are built. It is also widely used in the OR literature as a baseline reference algorithm against which other methods are compared.

The limitation of Clarke-Wright is that it is greedy. It makes locally good decisions at each step but cannot go back and correct early mistakes. This typically produces solutions within 5 to 15 percent above optimal.

OR-Tools with Guided Local Search

Google OR-Tools has become one of the most widely adopted open-source optimisation frameworks for industrial routing applications because it provides efficient implementations of routing algorithms capable of handling heterogeneous fleets, vehicle capacities and customer time windows. Consequently, OR-Tools was selected as the computational framework for the present study. We use it with two phases: `PATH_CHEAPEST_ARC` to build an initial solution and Guided Local Search to improve it.

Guided Local Search is a metaheuristic that escapes local optima by adding penalty terms to arcs that are used too frequently. This forces the solver to explore alternative routings and is documented in the metaheuristics literature to consistently produce better solutions than pure constructive heuristics.

We chose OR-Tools because it natively supports heterogeneous fleets through vehicle-indexed capacity dimensions, time windows through cumulative time constraints, and it can handle the time-dependent callback that applies $\mu(t)$ at each arc evaluation.

5. Comparison: Clarke-Wright vs OR-Tools

We ran both methods on Scenario S-99 (99 pallets, baseline).

Criterion	Clarke-Wright	OR-Tools GLS
Total distance	~308 km	215 km
Solution time	under 1 ms	under 10 sec
Routes generated	4	4
All stores served	Yes	Yes
Optimality guarantee	None	Near-optimal

OR-Tools achieves approximately 30 percent shorter total distance compared to Clarke-Wright. This is consistent with general findings in the VRP literature, where metaheuristic improvement phases regularly outperform pure constructive heuristics on medium-sized instances.

Clarke-Wright is still valuable because it runs instantly and produces an interpretable solution that shows clearly why certain stores are grouped together. OR-Tools is better for production use where solution quality matters more than speed or transparency.

The comparison demonstrates the trade-off between computational efficiency and solution quality. Clarke-Wright generates feasible routes almost instantaneously but sacrifices solution quality because early

greedy decisions cannot be revised. In contrast, OR-Tools requires longer computation time but consistently produces shorter delivery routes through Guided Local Search. For practical logistics planning, the improvement in route quality outweighs the modest increase in computational time.

6. What are the limitations of our model?

Simplified congestion model

Our $\mu(t)$ step function is a useful approximation but does not capture link-level traffic variation. The same multiplier is applied to all arcs at a given time, which does not reflect that the A2 motorway may be more congested than a local street. This is a recognized limitation of step-function congestion models in general: link-specific time-dependent functions would be more accurate but require detailed traffic sensor data that was not available to us.

No split deliveries

Constraint C1 requires each store to be visited exactly once by one vehicle. This means a store with 8 pallets must receive all 8 from the same truck. Allowing split deliveries could make Scenario S-170 feasible but would significantly increase the model complexity, as discussed by Toth and Vigo (2014).

Static demand

All three scenarios use fixed deterministic demands. In reality daily demand varies. A stochastic extension would use probability distributions for demand but is beyond the scope of this project.

Heuristic optimality gap

Clarke-Wright does not guarantee optimality. OR-Tools with GLS provides near-optimal solutions within the 10 second time limit but also cannot prove optimality. For a provably optimal solution an exact ILP solver such as Gurobi would be needed, which was not available to us.

Conclusion

The modelling decisions adopted in this project are grounded in established Vehicle Routing Problem literature and were selected to represent the operational characteristics of the Dr. Oetker–Lidl distribution network as realistically as possible. Beginning with the classical CVRP, the model was extended to incorporate heterogeneous vehicle capacities, customer time windows, and time-dependent travel times, resulting in a TDHVRPTW formulation. These extensions improve the realism of the optimisation model while maintaining computational tractability. Although certain simplifying assumptions were necessary, the model provides a practical decision-support framework for regional food distribution and establishes a solid foundation for future research involving dynamic traffic information, stochastic demand, and exact optimisation techniques.

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